



Techniques de sécurité informatique
420-4D5-MO

Travail pratique 2 – Partie 2

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Filtrage ACL (4,5 points)..... 33

Travail pratique 2 – Partie 2 (10%)

Filtrage ACL et WiFi sécurisé

Le travail pratique 2 comporte 3 parties :

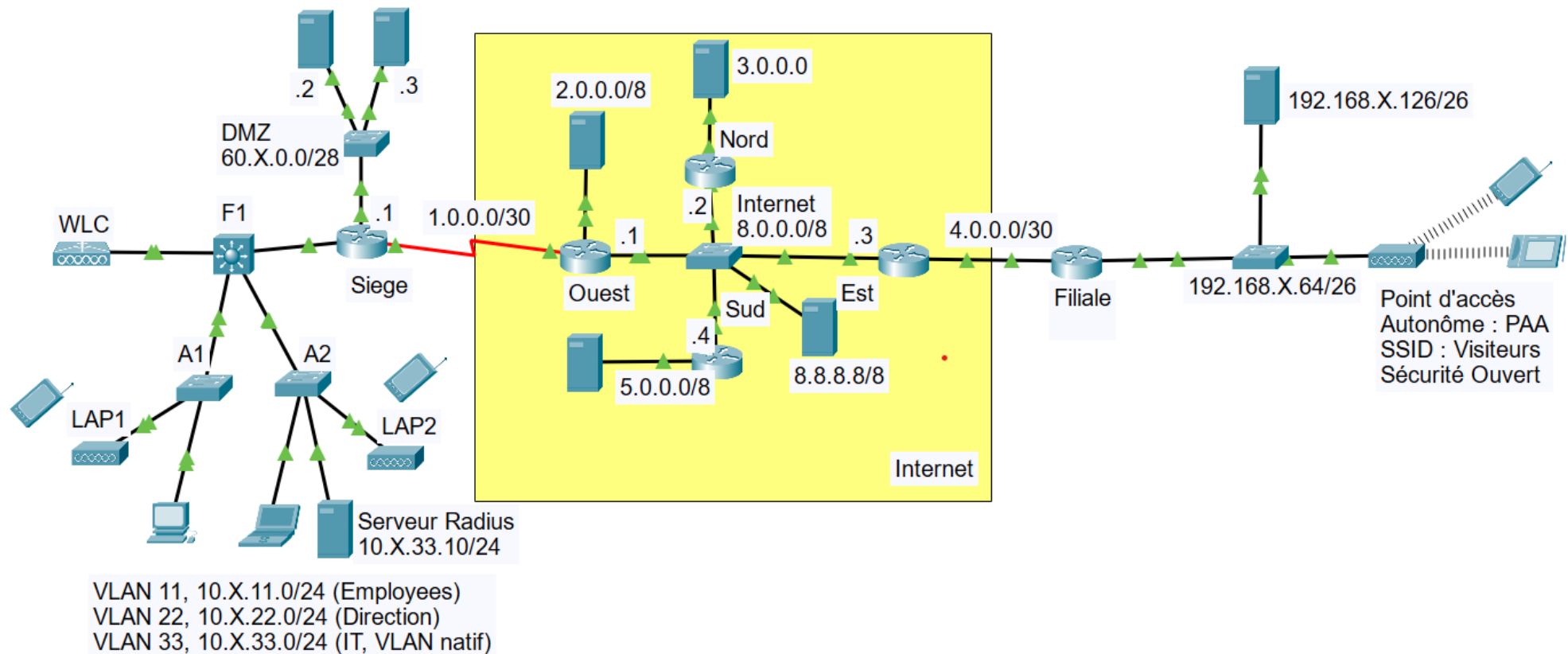
- Partie 1 : Attaque DOS et Authentification sécurisée (10 points)
- **Partie 2 : Filtrage ACL et WiFi sécurisé (10 points)**
- Partie 3 : Filtrage avec PfSense (10 points).

La validation en classe compte pour 50% de la note de chaque partie.

Partie 2 : Filtrage ACL et WiFi sécurisé (10 points)

La partie 2 du TP2 est à valider lors de la séance du jeudi 7 mai et les fichiers (TP2-SA.pkt et TP2-AA.pkt) sont à soumettre le même jour avant minuit.

Réalisez la maquette ci-après. Votre X est celui donné au labo1.



Adressage et routage (2 points)

- 1- (0,5) Dans 'Cisco Packettracer', chargez le fichier TP2-Partie2.pkt. Configurez le routage inter-VLAN ainsi que le service DHCP sur le routeur **Siege** pour tous les VLANs ?

A1:

```
Switch(config)#hostname A1
A1(config)#
A1(config)#vlan 11
A1(config-vlan)# name Employees
A1(config-vlan)#exit
A1(config)#vlan 22
A1(config-vlan)# name Direction
A1(config-vlan)#exit
A1(config)#vlan 33
A1(config-vlan)# name IT
A1(config-vlan)#exit
A1(config)#
A1(config)#interface fastEthernet 0/18
A1(config-if)# switchport mode trunk

A1(config-if)# switchport trunk native vlan 33
A1(config-if)# switchport trunk allowed vlan 11,22,33
A1(config-if)#exit
A1(config)#
A1(config)#interface fastEthernet 0/19
A1(config-if)# switchport mode access
A1(config-if)# switchport access vlan 11
A1(config-if)#exit
A1(config)#
A1(config)#interface fastEthernet 0/20
A1(config-if)# switchport mode access
A1(config-if)# switchport access vlan 33
```

```
Switch(config)#hostname A2
A2(config)#
A2(config)#vlan 11
A2(config-vlan)# name Employees
A2(config-vlan)#exit
A2(config)#vlan 22
A2(config-vlan)# name Direction
A2(config-vlan)#exit
A2(config)#vlan 33
A2(config-vlan)# name IT
A2(config-vlan)#exit
A2(config)#
A2(config)#interface fastEthernet 0/19
A2(config-if)# switchport mode trunk

A2(config-if)# switchport trunk native vlan 33
A2(config-if)# switchport trunk allowed vlan 11,22,33
A2(config-if)#exit
A2(config)#
A2(config)#interface fastEthernet 0/20
A2(config-if)# switchport mode access
A2(config-if)# switchport access vlan 22
A2(config-if)#exit
A2(config)#
A2(config)#interface fastEthernet 0/21
A2(config-if)# switchport mode access
A2(config-if)# switchport access vlan 33
A2(config-if)#exit
A2(config)#
A2(config)#interface fastEthernet 0/22
A2(config-if)# switchport mode access
A2(config-if)# switchport access vlan 33
```

A2:

```
Switch(config)#hostname F1
F1(config)#
F1(config)#vlan 11
F1(config-vlan)# name Employees
F1(config-vlan)#exit
F1(config)#vlan 22
F1(config-vlan)# name Direction
F1(config-vlan)#exit
F1(config)#vlan 33
F1(config-vlan)# name IT
F1(config-vlan)#exit
F1(config)#
F1(config)#interface fastEthernet 0/15
F1(config-if)# switchport trunk encapsulation dot1q
F1(config-if)# switchport mode trunk
F1(config-if)# switchport trunk native vlan 33
F1(config-if)# switchport trunk allowed vlan 11,22,33
F1(config-if)#exit
F1(config)#
F1(config)#interface fastEthernet 0/16
F1(config-if)# switchport trunk encapsulation dot1q
F1(config-if)# switchport mode trunk
F1(config-if)# switchport trunk native vlan 33
F1(config-if)# switchport trunk allowed vlan 11,22,33
F1(config-if)#exit
F1(config)#
F1(config)#interface gigabitEthernet 0/1
F1(config-if)# switchport trunk encapsulation dot1q
F1(config-if)# switchport mode trunk

F1(config-if)# switchport trunk native vlan 33
F1(config-if)# switchport trunk allowed vlan 11,22,33
F1(config-if)#exit
F1(config)#
F1(config)#interface gigabitEthernet 0/2
F1(config-if)# switchport mode access
F1(config-if)# switchport access vlan 33
```

F1: 11, 22, 33

```

Siege(config)#interface gigabitEthernet 0/0
Siege(config-if)# no shutdown
Siege(config-if)#exit
Siege(config)#
Siege(config)#interface gigabitEthernet 0/0.11
Siege(config-subif)# encapsulation dot1Q 11
Siege(config-subif)# ip address 10.64.11.1 255.255.255.0
Siege(config-subif)#exit
Siege(config)#
Siege(config)#interface gigabitEthernet 0/0.22
Siege(config-subif)# encapsulation dot1Q 22
Siege(config-subif)# ip address 10.64.22.1 255.255.255.0
Siege(config-subif)#exit
Siege(config)#
Siege(config)#interface gigabitEthernet 0/0.33
Siege(config-subif)# encapsulation dot1Q 33 native
Siege(config-subif)# ip address 10.64.33.1 255.255.255.0
Siege(config-subif)#exit
Siege(config)#
Siege(config)#ip dhcp excluded-address 10.64.11.1 10.64.11.10
Siege(config)#ip dhcp excluded-address 10.64.22.1 10.64.22.10
Siege(config)#ip dhcp excluded-address 10.64.33.1 10.64.33.20
Siege(config)#
Siege(config)#ip dhcp pool POOL_EMPLOYEES
Siege(dhcp-config)# network 10.64.11.0 255.255.255.0
Siege(dhcp-config)# default-router 10.64.11.1
Siege(dhcp-config)# dns-server 60.64.0.2
Siege(dhcp-config)#exit
Siege(config)#
Siege(config)#ip dhcp pool POOL_DIRECTION
Siege(dhcp-config)# network 10.64.22.0 255.255.255.0
Siege(dhcp-config)# default-router 10.64.22.1
Siege(dhcp-config)# dns-server 60.64.0.2
Siege(dhcp-config)#exit
Siege(config)#
Siege(config)#ip dhcp pool POOL_IT
Siege(dhcp-config)# network 10.64.33.0 255.255.255.0
Siege(dhcp-config)# default-router 10.64.33.1
Siege(dhcp-config)# dns-server 60.64.0.2

```

SIEGE:

2- (0,5) Configurez le routage OSPF sur les routeurs **Ouest, Est, Nord et Sud** ? Sécurisez le routage OSPF avec la clé partagée CmMO2026 ?

```

Siege(config)#interface serial 0/0/0
Siege(config-if)# ip address 1.64.0.2 255.255.255.252
Siege(config-if)# encapsulation hdlc
Siege(config-if)# clock rate 64000
Siege(config-if)# no shutdown

```

Siege :

```
Quest(config)#interface serial 0/0/0
Quest(config-if)# ip address 1.64.0.1 255.255.255.252
Quest(config-if)# encapsulation hdlc
Quest(config-if)# clock rate 64000
This command applies only to DCE interfaces
Quest(config-if)# no shutdown
Quest(config-if)#exit
Quest(config)#
Quest(config)#interface gigabitEthernet 0/0
Quest(config-if)# ip address 100.64.0.1 255.255.255.0
Quest(config-if)# no shutdown
Quest(config-if)#exit
Quest(config)#
Quest(config)#interface gigabitEthernet 0/1
Quest(config-if)# ip address 2.64.0.1 255.255.0.0
Quest(config-if)# no shutdown
Quest(config-if)#exit
Quest(config)#
Quest(config)#router ospf 1
Quest(config-router)# router-id 1.1.1.1
Quest(config-router)# network 1.64.0.0 0.0.0.3 area 0
Quest(config-router)# network 100.64.0.0 0.0.0.255 area 0
Quest(config-router)# network 2.64.0.0 0.0.255.255 area 0
Quest(config-router)# area 0 authentication message-digest
Quest(config-router)#exit
Quest(config)#
Quest(config)#interface serial 0/0/0
Quest(config-if)# ip ospf message-digest-key 1 md5 CmMO2026
Quest(config-if)#exit
Quest(config)#
Quest(config)#interface gigabitEthernet 0/0
Quest(config-if)# ip ospf message-digest-key 1 md5 CmMO2026
Quest(config-if)#exit
Quest(config)#
Quest(config)#interface gigabitEthernet 0/1
Quest(config-if)# ip ospf message-digest-key 1 md5 CmMO2026
```

OUEST :

```

Nord(config)#interface gigabitEthernet 0/1
Nord(config-if)# ip address 100.64.0.2 255.255.255.0
Nord(config-if)# no shutdown
Nord(config-if)#exit
Nord(config)#
Nord(config)#interface gigabitEthernet 0/0
Nord(config-if)# ip address 3.64.0.1 255.255.0.0
Nord(config-if)# no shutdown
Nord(config-if)#exit
Nord(config)#
Nord(config)#router ospf 1
Nord(config-router)# router-id 2.2.2.2
Nord(config-router)# network 100.64.0.0 0.0.0.255 area 0
Nord(config-router)# network 3.64.0.0 0.0.255.255 area 0
Nord(config-router)# area 0 authentication message-digest
Nord(config-router)#exit
Nord(config)#
Nord(config)#interface gigabitEthernet 0/1
Nord(config-if)# ip ospf message-digest-key 1 md5 CmMO2026
Nord(config-if)#exit
Nord(config)#
Nord(config)#interface gigabitEthernet 0/0
Nord(config-if)# ip ospf message-digest-key 1 md5 CmMO2026
Nord(config-if)#exit

```

Nord :

```

Sud(config)#
Sud(config)#interface gigabitEthernet 0/0
Sud(config-if)# ip address 100.64.0.3 255.255.255.0
Sud(config-if)# no shutdown
Sud(config-if)#exit
Sud(config)#
Sud(config)#interface gigabitEthernet 0/1
Sud(config-if)# ip address 5.64.0.1 255.255.0.0
Sud(config-if)# no shutdown
Sud(config-if)#exit
Sud(config)#
Sud(config)#router ospf 1
Sud(config-router)# router-id 3.3.3.3
Sud(config-router)# network 100.64.0.0 0.0.0.255 area 0
Sud(config-router)# network 5.64.0.0 0.0.255.255 area 0
Sud(config-router)# network 8.0.0.0 0.255.255.255 area 0
Sud(config-router)# area 0 authentication message-digest
Sud(config-router)#exit
Sud(config)#
Sud(config)#interface gigabitEthernet 0/0
Sud(config-if)# ip ospf message-digest-key 1 md5 CmMO2026
Sud(config-if)#exit
Sud(config)#
Sud(config)#interface gigabitEthernet 0/1
Sud(config-if)# ip ospf message-digest-key 1 md5 CmMO2026

```

Sud :

```
Est(config)#interface gigabitEthernet 0/1
Est(config-if)# ip address 100.64.0.4 255.255.255.0
Est(config-if)# no shutdown
Est(config-if)#exit
Est(config)#
Est(config)#interface gigabitEthernet 0/0
Est(config-if)# ip address 4.64.0.1 255.255.255.252
Est(config-if)# no shutdown
Est(config-if)#exit
Est(config)#
Est(config)#router ospf 1
Est(config-router)# router-id 4.4.4.4
Est(config-router)# network 100.64.0.0 0.0.0.255 area 0
Est(config-router)# network 4.64.0.0 0.0.0.3 area 0
Est(config-router)# area 0 authentication message-digest
Est(config-router)#exit
Est(config)#
Est(config)#interface gigabitEthernet 0/1
Est(config-if)# ip ospf message-digest-key 1 md5 CmMO2026
Est(config-if)#exit
Est(config)#
Est(config)#interface gigabitEthernet 0/0
Est(config-if)# ip ospf message-digest-key 1 md5 CmMO2026
```

EST :

- 3- (0,5) Configurez une route statique par défaut ainsi que le protocole NAT sur le routeur **Siege**, autorisez les VLANs IT et Direction à sortir vers Internet ?
- Siege :

```
Siege(config)#interface gigabitEthernet 0/1
Siege(config-if)# ip address 60.64.0.1 255.255.255.240
Siege(config-if)# ip nat inside
Siege(config-if)# no shutdown
Siege(config-if)#exit
Siege(config)#
Siege(config)#ip route 0.0.0.0 0.0.0.0 1.64.0.1
Siege(config)#
Siege(config)#access-list 1 permit 10.64.22.0 0.0.0.255
Siege(config)#access-list 1 permit 10.64.33.0 0.0.0.255
Siege(config)#
Siege(config)#interface gigabitEthernet 0/0.11
Siege(config-subif)# ip nat inside
Siege(config-subif)#exit
Siege(config)#
Siege(config)#interface gigabitEthernet 0/0.22
Siege(config-subif)# ip nat inside
Siege(config-subif)#exit
Siege(config)#
Siege(config)#interface gigabitEthernet 0/0.33
Siege(config-subif)# ip nat inside
Siege(config-subif)#exit
Siege(config)#
Siege(config)#interface serial 0/0/0
Siege(config-if)# ip nat outside
Siege(config-if)#exit
Siege(config)#
Siege(config)#ip nat inside source list 1 interface serial 0/0/0 overload
```

- 4- (0,25) Configurez une route statique par défaut ainsi que le protocole NAT sur le routeur **Filiale** ?

```

Filiale(config)#interface gigabitEthernet 0/0
Filiale(config-if)# ip address 4.64.0.2 255.255.255.252
Filiale(config-if)# ip nat outside
Filiale(config-if)# no shutdown
Filiale(config-if)#exit
Filiale(config)#
Filiale(config)#interface gigabitEthernet 0/1
Filiale(config-if)# ip address 192.168.64.65 255.255.255.192
Filiale(config-if)# ip nat inside
Filiale(config-if)# no shutdown
Filiale(config-if)#exit
Filiale(config)#
Filiale(config)#ip route 0.0.0.0 0.0.0.0 4.64.0.1
Filiale(config)#
Filiale(config)#access-list 1 permit 192.168.64.64 0.0.0.63
Filiale(config)#ip nat inside source list 1 interface gigabitEthernet 0/0 overload
Filiale(config)#
Filiale(config)#ip dhcp excluded-address 192.168.64.65 192.168.64.75
Filiale(config)#
Filiale(config)#ip dhcp pool POOL_FILIALE
Filiale(dhcp-config)# network 192.168.64.64 255.255.255.192
Filiale(dhcp-config)# default-router 192.168.64.65
Filiale(dhcp-config)# dns-server 60.64.0.2

```

- 5- (0,25) Configurez le service DNS (votre-nom.ca) sur la machine 60.X.0.2/28 de la DMZ et le service WEB (www.votre-nom.ca) sur la machine 60.X.0.3/28 ?

Server7

Physical Config Services **Desktop** Programming Attributes

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 60.64.0.2

Subnet Mask 255.255.255.240

Default Gateway 60.64.0.1

DNS Server 60.64.0.2

Server7 DNS :

DNS

DNS Service

☒ On

☐ Off

Resource Records

Name

Type

A Record

Address

Add

Save

No.	Name	Type	
0	romeo.ca	A Record	60.64.0.3
1	www.romeo.com	A Record	60.64.0.3

Server3

PhysicalConfigServicesDesktopProgrammingAttributes

Web Browser

<>URLhttp://www.romeo.com

Cisco Packet Tracer

Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.

Quick Links:
[A small page](#)
[Copyrights](#)
[Image page](#)
[Image](#)

Server6 WEB :

The screenshot shows the 'Server6' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted in blue. It shows the 'Static' radio button selected, with the following values: IPv4 Address: 60.64.0.3, Subnet Mask: 255.255.255.240, Default Gateway: 60.64.0.1, and DNS Server: 60.64.0.2.

Field	Value
IPv4 Address	60.64.0.3
Subnet Mask	255.255.255.240
Default Gateway	60.64.0.1
DNS Server	60.64.0.2

The screenshot shows the 'Server6' configuration window with the 'Services' tab selected. The 'HTTP' service is highlighted in the left sidebar and is currently 'On'. Below the service status, there is a 'File Manager' table listing several files.

File Name	Edit
1 copyrights.html	(edit)
2 cscoptlogo177x111.jpg	
3 helloworld.html	(edit)
4 image.html	(edit)
5 index.html	(edit)

6- Testez les communications depuis tous les VLANs du siège et de la filiale vers la DMZ et Internet ?

```

C:\>ping 10.64.11.1
Pinging 10.64.11.1 with 32 bytes of data:

Reply from 10.64.11.1: bytes=32 time<1ms TTL=255
Reply from 10.64.11.1: bytes=32 time<1ms TTL=255
Reply from 10.64.11.1: bytes=32 time<1ms TTL=255
Reply from 10.64.11.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.64.11.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 60.64.0.2
Pinging 60.64.0.2 with 32 bytes of data:

Request timed out.
Reply from 60.64.0.2: bytes=32 time=1ms TTL=127
Reply from 60.64.0.2: bytes=32 time=1ms TTL=127
Reply from 60.64.0.2: bytes=32 time<1ms TTL=127

Ping statistics for 60.64.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 60.64.0.3
Pinging 60.64.0.3 with 32 bytes of data:

Request timed out.
Reply from 60.64.0.3: bytes=32 time<1ms TTL=127
Reply from 60.64.0.3: bytes=32 time<1ms TTL=127
Reply from 60.64.0.3: bytes=32 time<1ms TTL=127

Ping statistics for 60.64.0.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

Sur PC0 VLAN11 :

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 10.64.11.1: Destination host unreachable.
Reply from 10.64.11.1: Destination host unreachable.
Reply from 10.64.11.1: Destination host unreachable.
Reply from 10.64.11.1: Destination host unreachable.

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

Laptop0 VLAN 22 :

```
C:\>ping 10.64.22.1
```

```
Pinging 10.64.22.1 with 32 bytes of data:
```

```
Reply from 10.64.22.1: bytes=32 time<1ms TTL=255  
Reply from 10.64.22.1: bytes=32 time<1ms TTL=255  
Reply from 10.64.22.1: bytes=32 time<1ms TTL=255  
Reply from 10.64.22.1: bytes=32 time<1ms TTL=255
```

```
Ping statistics for 10.64.22.1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 60.64.0.2
```

```
Pinging 60.64.0.2 with 32 bytes of data:
```

```
Reply from 60.64.0.2: bytes=32 time<1ms TTL=127  
Reply from 60.64.0.2: bytes=32 time<1ms TTL=127  
Reply from 60.64.0.2: bytes=32 time<1ms TTL=127  
Reply from 60.64.0.2: bytes=32 time<1ms TTL=127
```

```
Ping statistics for 60.64.0.2:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 60.64.0.3
```

```
Pinging 60.64.0.3 with 32 bytes of data:
```

```
Reply from 60.64.0.3: bytes=32 time<1ms TTL=127  
Reply from 60.64.0.3: bytes=32 time<1ms TTL=127  
Reply from 60.64.0.3: bytes=32 time<1ms TTL=127  
Reply from 60.64.0.3: bytes=32 time<1ms TTL=127
```

```
Ping statistics for 60.64.0.3:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 8.8.8.8
```

```
Pinging 8.8.8.8 with 32 bytes of data:
```

```
Reply from 8.8.8.8: bytes=32 time=20ms TTL=125
```

```
Reply from 8.8.8.8: bytes=32 time=15ms TTL=125
```

```
Reply from 8.8.8.8: bytes=32 time=14ms TTL=125
```

```
Reply from 8.8.8.8: bytes=32 time=16ms TTL=125
```

```
Ping statistics for 8.8.8.8:
```

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
Minimum = 14ms, Maximum = 20ms, Average = 16ms
```

Server2

Physical Config Services **Desktop** Programming Attributes

IP Configuration

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 10.64.33.10

Subnet Mask 255.255.255.0

Default Gateway 10.64.3.1

DNS Server 60.64.0.2

Server2 VLAN 33 :

```
C:\>ping 10.64.33.1
```

```
Pinging 10.64.33.1 with 32 bytes of data:
```

```
Reply from 10.64.33.1: bytes=32 time<1ms TTL=255  
Reply from 10.64.33.1: bytes=32 time<1ms TTL=255  
Reply from 10.64.33.1: bytes=32 time<1ms TTL=255  
Reply from 10.64.33.1: bytes=32 time<1ms TTL=255
```

```
Ping statistics for 10.64.33.1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 60.64.0.2
```

```
Pinging 60.64.0.2 with 32 bytes of data:
```

```
Reply from 60.64.0.2: bytes=32 time<1ms TTL=127  
Reply from 60.64.0.2: bytes=32 time<1ms TTL=127  
Reply from 60.64.0.2: bytes=32 time=10ms TTL=127  
Reply from 60.64.0.2: bytes=32 time<1ms TTL=127
```

```
Ping statistics for 60.64.0.2:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

```
C:\>ping 60.64.0.3
```

```
Pinging 60.64.0.3 with 32 bytes of data:
```

```
Reply from 60.64.0.3: bytes=32 time<1ms TTL=127  
Reply from 60.64.0.3: bytes=32 time<1ms TTL=127  
Reply from 60.64.0.3: bytes=32 time<1ms TTL=127  
Reply from 60.64.0.3: bytes=32 time<1ms TTL=127
```

```
Ping statistics for 60.64.0.3:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 8.8.8.8
```

```
Pinging 8.8.8.8 with 32 bytes of data:
```

```
Reply from 8.8.8.8: bytes=32 time=2ms TTL=125  
Reply from 8.8.8.8: bytes=32 time=23ms TTL=125  
Reply from 8.8.8.8: bytes=32 time=1ms TTL=125  
Reply from 8.8.8.8: bytes=32 time=10ms TTL=125
```

```
C:\>ping 192.168.64.65

Pinging 192.168.64.65 with 32 bytes of data:

Reply from 192.168.64.65: bytes=32 time=20ms TTL=255
Reply from 192.168.64.65: bytes=32 time=29ms TTL=255
Reply from 192.168.64.65: bytes=32 time=20ms TTL=255

Ping statistics for 192.168.64.65:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 20ms, Maximum = 29ms, Average = 23ms

Control-C
^C
C:\>ping 60.64.0.2

Pinging 60.64.0.2 with 32 bytes of data:

Reply from 60.64.0.2: bytes=32 time=13ms TTL=124
Reply from 60.64.0.2: bytes=32 time=9ms TTL=124
Reply from 60.64.0.2: bytes=32 time=5ms TTL=124
Reply from 60.64.0.2: bytes=32 time=23ms TTL=124

Ping statistics for 60.64.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 23ms, Average = 12ms

C:\>
C:\>ping 60.64.0.3

Pinging 60.64.0.3 with 32 bytes of data:

Reply from 60.64.0.3: bytes=32 time=17ms TTL=124
Reply from 60.64.0.3: bytes=32 time=11ms TTL=124
Reply from 60.64.0.3: bytes=32 time=8ms TTL=124
Reply from 60.64.0.3: bytes=32 time=20ms TTL=124

Ping statistics for 60.64.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 20ms, Average = 14ms
```

Smartphone0 – Filiale :

```

C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=58ms TTL=125
Reply from 8.8.8.8: bytes=32 time=12ms TTL=125
Reply from 8.8.8.8: bytes=32 time=13ms TTL=125
Reply from 8.8.8.8: bytes=32 time=26ms TTL=125

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 12ms, Maximum = 58ms, Average = 27ms

C:\>

```

Réseaux sans-fil (3,5 points)

7- (0,5) Configurez le point d'accès autonome **PAA** avec les informations de sécurité indiquées sur la maquette ci-dessus ?

Port 1	
Port Status	☒ On
SSID	Visiteurs
5 GHz Channel	112
Coverage Range (meters)	140.00
Authentication	☐ WEP ☒ WPA2-PSK
Encryption Type	AES
WEP Key	
PSK Pass Phrase	
User ID	
Password	CmMO2026

8- Connectez un smartphone au point d'accès autonome **PAA** ? puis testez la connexion vers la DMZ et Internet ?

Wireless0

Port Status ☒ On

Bandwidth 300 Mbps

MAC Address 000A.4167.D6CC

SSID Visiteurs

Authentication

☐ Disabled

☐ WPA-PSK

☐ WPA

☐ 802.1X

☐ WEP

☒ WPA2-PSK

☐ WPA2

Method:

WEK Key

PSK Pass Phrase

User ID CmlMO2026

Password

MD5

User Name

Password

AES

Encryption Type

IP Configuration

☒ DHCP 

☐ Static

IPv4 Address

Subnet Mask

192.168.64.76 

255.255.255.192

C:\>ipconfig

Wireless0 Connection: (default port)

Connection-specific DNS Suffix...
Link-local IPv6 Address..... FE80::20A:41FF:FE67:D6CC
IPv6 Address..... ::
IPv4 Address..... 192.168.64.76 
Subnet Mask..... 255.255.255.192
Default Gateway..... ::
192.168.64.65

```
C:\>ping 192.168.64.65
Pinging 192.168.64.65 with 32 bytes of data:

Reply from 192.168.64.65: bytes=32 time=60ms TTL=255
Reply from 192.168.64.65: bytes=32 time=13ms TTL=255
Reply from 192.168.64.65: bytes=32 time=17ms TTL=255
Reply from 192.168.64.65: bytes=32 time=26ms TTL=255

Ping statistics for 192.168.64.65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 13ms, Maximum = 60ms, Average = 29ms

C:\>ping 60.64.0.2
Pinging 60.64.0.2 with 32 bytes of data:

Reply from 60.64.0.2: bytes=32 time=23ms TTL=124
Reply from 60.64.0.2: bytes=32 time=7ms TTL=124
Reply from 60.64.0.2: bytes=32 time=10ms TTL=124
Reply from 60.64.0.2: bytes=32 time=14ms TTL=124

Ping statistics for 60.64.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 23ms, Average = 13ms

C:\>ping 60.64.0.3
Pinging 60.64.0.3 with 32 bytes of data:

Reply from 60.64.0.3: bytes=32 time=24ms TTL=124
Reply from 60.64.0.3: bytes=32 time=7ms TTL=124
Reply from 60.64.0.3: bytes=32 time=10ms TTL=124
Reply from 60.64.0.3: bytes=32 time=10ms TTL=124

Ping statistics for 60.64.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 24ms, Average = 12ms
```

- 9- Sur le WLC du réseau Siege, configurez l'accès avec le compte : Admin et le mot de passe : CmM2026

Management

IP Configuration

IPv4 Address

10.64.33.2

Subnet Mask

255.255.255.0

Default Gateway

10.64.33.1

DNS Server

60.64.0.2

Web Browser

URL http://10.64.33.2



Cisco 3500 Series Wireless LAN Controller

Welcome! Please start by creating an admin account.

 Admin





Start



Cisco 3500 Series Wireless LAN Controller

1 Set Up Your Controller



System Name ?

Country ?

Date & Time 

Timezone ?

NTP Server ?

Management IP Address ?

Subnet Mask

Default Gateway

Management VLAN ID ?

Back

Next

- 10-Sur le WLC, configurez deux WLANs sécurisés :
- a. (1 point) WLAN : **Direction** avec WPA2 personnel, clé : Direction202!

☐ WLAN ID	Type	Profile Name	WLAN SSID	Admin Status	Security Policies
☐ 1	WLAN	Direction	Direction	Enabled	[WPA2][Auth(PSK)]
☐ 2	WLAN	Employees	Employees	Enabled	[WPA2][Auth(802.1X)]

- b. (2 points) WLAN : **Employees** avec WPA2 entreprise, le serveur radius sera configuré sur la machine 10.X.33.10/24 (clé radius : AAAkey202# et le port d'écoute : 1645 ?

```
F1(config)#interface gigabitEthernet 0/2
F1(config-if)# switchport trunk encapsulation dot1q
F1(config-if)# switchport mode trunk

F1(config-if)# switchport trunk native vlan 33
F1(config-if)# switchport trunk allowed vlan 11,22,33
F1(config-if)#exit
```

Server2

Physical Config Services Desktop Programming Attributes

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 10.64.33.10

Subnet Mask 255.255.255.0

Default Gateway 10.64.33.1

DNS Server 60.64.0.2

Server2

Physical Config Services Desktop Programming Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

AAA

Service ☐ On ☒ Off Radius Port

Network Configuration

Client Name Client IP

Secret ServerType Radius

	Client Name	Client IP	Server Type	
1	WLC	10.64.33.2	Radius	AAAkey202#

User Setup

Username Password

	Username	
1	emp1	CmMO2026

Interfaces

Interface Name	VLAN Identifier	IP Address	Interface Type	Dynamic AP Management	IPv6 Address	
Direction	22	10.64.22.2	Dynamic	Disabled		Remove
Employees	11	10.64.11.2	Dynamic	Disabled		Remove
management	33	10.64.33.2	Static	Enabled	::/128	
virtual	N/A	192.0.2.1	Static	Not Supported		

RADIUS Authentication Servers

Auth Called Station ID Type

Use AES Key Wrap ☐ (Designed for FIPS customers and requires a key wrap compliant RADIUS server)

MAC Delimiter

Framed MTU

Network User	Management	Server Index	Server Address(Ipv4/Ipv6)	Port	IPSec	Admin Status	
☒	☒	1	10.64.33.10	1645	Disabled	Enabled	Remove

☐ WLAN ID	Type	Profile Name	WLAN SSID	Admin Status	Security Policies
☐ 1	WLAN	Direction	Direction	Enabled	[WPA2][Auth(PSK)]
☐ 2	WLAN	Employees	Employees	Enabled	[WPA2][Auth(802.1X)]

```

A1(config)#interface fastEthernet 0/20
A1(config-if)# switchport mode trunk

A1(config-if)# switchport trunk native vlan 33
A1(config-if)# switchport trunk allowed vlan 11,22,33
A1(config-if)#exit
A1(config)#
A1(config)#end
A1#write memory

A2(config)#interface fastEthernet 0/22
A2(config-if)# switchport mode trunk

A2(config-if)# switchport trunk native vlan 33
A2(config-if)# switchport trunk allowed vlan 11,22,33

```

- c. Connectez un client sans-fil sur le WLAN **Direction** et un deuxième client sans-fil sur le WLAN **Employees** ?
Laptop0 :

SSID

Authentication
☐ Disabled
☐ WPA-PSK
☐ WPA
☐ 802.1X

☐ WEP
☒ WPA2-PSK
☐ WPA2

Method:

Encryption Type

Direction

WEP Key

PSK Pass Phrase

User ID

Password

MD5

User Name

Password

AES

Direction202!

IP Configuration

☒ DHCP

☐ Static

IPv4 Address

Subnet Mask

10.64.22.13

255.255.255.0

C:\>ping 10.64.22.1

Pinging 10.64.22.1 with 32 bytes of data:

Request timed out.

Reply from 10.64.22.1: bytes=32 time=52ms TTL=255

Reply from 10.64.22.1: bytes=32 time=24ms TTL=255

Reply from 10.64.22.1: bytes=32 time=17ms TTL=255

Ping statistics for 10.64.22.1:

Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),

Approximate round trip times in milli-seconds:

Minimum = 17ms, Maximum = 52ms, Average = 31ms

Smartphone2 :

SSID

Authentication

- ☐ Disabled
- ☐ WPA-PSK
- ☐ WPA
- ☐ 802.1X

Encryption Type

IP Configuration

- ☒ DHCP
- ☐ Static

IPv4 Address

Subnet Mask

Employees

WEP Key

PSK Pass Phrase

User ID

Password

MDS

User Name

Password

AES

10.64.11.12

255.255.255.0

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.64.11.1

Pinging 10.64.11.1 with 32 bytes of data:

Reply from 10.64.11.1: bytes=32 time=40ms TTL=255
Reply from 10.64.11.1: bytes=32 time=17ms TTL=255
Reply from 10.64.11.1: bytes=32 time=17ms TTL=255

Ping statistics for 10.64.11.1:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 17ms, Maximum = 40ms, Average = 24ms

Control-C
^C
C:\>ping 10.64.11.2

Pinging 10.64.11.2 with 32 bytes of data:

Request timed out.
Reply from 10.64.11.2: bytes=32 time=21ms TTL=255
Reply from 10.64.11.2: bytes=32 time=9ms TTL=255
Reply from 10.64.11.2: bytes=32 time=11ms TTL=255

Ping statistics for 10.64.11.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 9ms, Maximum = 21ms, Average = 13ms
```

- d. Testez les communications entre les clients sans-fil gérés par le WLC ?
De Laptop0 à Smartphone2 :

```
C:\>ping 10.64.11.12

Pinging 10.64.11.12 with 32 bytes of data:

Reply from 10.64.11.12: bytes=32 time=60ms TTL=127
Reply from 10.64.11.12: bytes=32 time=21ms TTL=127
Reply from 10.64.11.12: bytes=32 time=25ms TTL=127
Reply from 10.64.11.12: bytes=32 time=40ms TTL=127

Ping statistics for 10.64.11.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 21ms, Maximum = 60ms, Average = 36ms
```

```
C:\>ping 10.64.22.13

Pinging 10.64.22.13 with 32 bytes of data:

Reply from 10.64.22.13: bytes=32 time=86ms TTL=127
Reply from 10.64.22.13: bytes=32 time=35ms TTL=127

Ping statistics for 10.64.22.13:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 35ms, Maximum = 86ms, Average = 60ms
```

De Smartphone2 à laptop :

- e. Testez les communications entre les clients sans-fil du réseau Siege vers la DMZ et vers Internet ?

```

C:\>ping 60.64.0.2
Pinging 60.64.0.2 with 32 bytes of data:

Request timed out.
Reply from 60.64.0.2: bytes=32 time=8ms TTL=127
Reply from 60.64.0.2: bytes=32 time=21ms TTL=127
Reply from 60.64.0.2: bytes=32 time=11ms TTL=127

Ping statistics for 60.64.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 21ms, Average = 13ms

C:\>ping 60.64.0.3
Pinging 60.64.0.3 with 32 bytes of data:

Request timed out.
Reply from 60.64.0.3: bytes=32 time=10ms TTL=127
Reply from 60.64.0.3: bytes=32 time=21ms TTL=127
Reply from 60.64.0.3: bytes=32 time=8ms TTL=127

Ping statistics for 60.64.0.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 21ms, Average = 13ms

```

A partir de smartphone2 - Employees :

```

C:\>Ping 8.8.8.8
Pinging 8.8.8.8 with 32 bytes of data:

Reply from 10.64.11.1: Destination host unreachable.
Reply from 10.64.11.1: Destination host unreachable.
Reply from 10.64.11.1: Destination host unreachable.
Reply from 10.64.11.1: Destination host unreachable.

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

Ping 8.8.8.8 :

```

C:\>ping 60.64.0.2

Pinging 60.64.0.2 with 32 bytes of data:

Reply from 60.64.0.2: bytes=32 time=56ms TTL=127
Reply from 60.64.0.2: bytes=32 time=8ms TTL=127
Reply from 60.64.0.2: bytes=32 time=27ms TTL=127
Reply from 60.64.0.2: bytes=32 time=9ms TTL=127

Ping statistics for 60.64.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 8ms, Maximum = 56ms, Average = 25ms

C:\>ping 60.64.0.3

Pinging 60.64.0.3 with 32 bytes of data:

Reply from 60.64.0.3: bytes=32 time=23ms TTL=127
Reply from 60.64.0.3: bytes=32 time=20ms TTL=127
Reply from 60.64.0.3: bytes=32 time=7ms TTL=127
Reply from 60.64.0.3: bytes=32 time=19ms TTL=127

Ping statistics for 60.64.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 7ms, Maximum = 23ms, Average = 17ms

```

A partir de laptop0 - Direction :

```

C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=17ms TTL=125
Reply from 8.8.8.8: bytes=32 time=18ms TTL=125
Reply from 8.8.8.8: bytes=32 time=14ms TTL=125
Reply from 8.8.8.8: bytes=32 time=14ms TTL=125

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 14ms, Maximum = 18ms, Average = 15ms

```

Ping 8.8.8.8 :

11- Sauvegardez votre maquette sous deux noms différents : **TP2-SA.pkt** (SA : Sans ACLs) et **TP2-AA.pkt** (AA : Avec ACLs)?

Filtrage ACL (4,5 points)

Continuez à travailler sur la maquette **TP2-AA.pkt**. Laissez la maquette sans ACLs (TP2-SA.pkt) intacte pour la validation avec votre enseignant.

12- (0,5) Autorisez sur le routeur Siege le service SSH pour le VLAN IT uniquement ?

```

Siege>enable
Siege#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Siege(config)#
Siege(config)#ip domain-name tp2.local
Siege(config)#username Admin privilege 15 secret CmM2026
Siege(config)#
Siege(config)#crypto key generate rsa modulus 1024
^
% Invalid input detected at '^' marker.

Siege(config)#ip ssh version 2
Please create RSA keys (of at least 768 bits size) to enable SSH v2.
Siege(config)#
Siege(config)#ip access-list standard SSH_IT
Siege(config-std-nacl)# permit 10.64.33.0 0.0.0.255
Siege(config-std-nacl)# deny any
Siege(config-std-nacl)#exit
Siege(config)#
Siege(config)#line vty 0 4
Siege(config-line)# login local
Siege(config-line)# transport input ssh
Siege(config-line)# access-class SSH_IT in
Siege(config-line)#exit
Siege(config)#
Siege(config)#end
Siege#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Siege#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Siege(config)#crypto key generate rsa
The name for the keys will be: Siege.tp2.local
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

```

Sur le routeur siege :

```

C:\>ssh -l Admin 10.64.33.1
Password:

```

Test du ssh : **Siege#**

13- (0,5) Configurez le routeur Siege pour bloquer les attaques de type brute force sur SSH ?

```

-----
Siege#enable
Siege#configure terminal
Enter configuration commands, one per line.  End with (
Siege(config)#
Siege(config)#login block-for 120 attempts 3 within 60
Siege(config)#login on-failure log
Siege(config)#login on-success log
Siege(config)#
Siege(config)#end
Siege#write memory

```

14- (1 point) Écrivez une ACL qui autorise tous les services pour le VLAN IT ?

```

Siege(config)#ip access-list extended ACL_IT
Siege(config-ext-nacl)# permit udp any eq bootpc any eq bootps
Siege(config-ext-nacl)# permit ip 10.64.33.0 0.0.0.255 any
Siege(config-ext-nacl)# deny ip any any
Siege(config-ext-nacl)#exit
Siege(config)#
Siege(config)#interface gigabitEthernet 0/0.33
Siege(config-subif)# ip access-group ACL_IT in
Siege(config-subif)#exit
Siege(config)#
Siege(config)#end
Siege#write memory

```

15- (1 point) Écrivez une ACL qui autorise les services WEB et DNS vers la DMZ et Internet pour le VLAN **Direction** ?

```

----
Siege#enable
Siege#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Siege(config)#
Siege(config)#ip access-list extended ACL_DIRECTION
Siege(config-ext-nacl)# permit udp any eq bootpc any eq bootps
Siege(config-ext-nacl)# permit udp 10.64.22.0 0.0.0.255 host 60.64.0.2 eq domain
Siege(config-ext-nacl)# permit tcp 10.64.22.0 0.0.0.255 host 60.64.0.2 eq domain
Siege(config-ext-nacl)# permit tcp 10.64.22.0 0.0.0.255 host 60.64.0.3 eq www
Siege(config-ext-nacl)# permit tcp 10.64.22.0 0.0.0.255 host 60.64.0.3 eq 443
Siege(config-ext-nacl)# permit udp 10.64.22.0 0.0.0.255 any eq domain
Siege(config-ext-nacl)# permit tcp 10.64.22.0 0.0.0.255 any eq www
Siege(config-ext-nacl)# permit tcp 10.64.22.0 0.0.0.255 any eq 443
Siege(config-ext-nacl)# permit icmp 10.64.22.0 0.0.0.255 host 60.64.0.2 echo
Siege(config-ext-nacl)# permit icmp 10.64.22.0 0.0.0.255 host 60.64.0.3 echo
Siege(config-ext-nacl)# permit icmp 10.64.22.0 0.0.0.255 host 8.8.8.8 echo
Siege(config-ext-nacl)# deny ip any any
Siege(config-ext-nacl)#exit
Siege(config)#
Siege(config)#interface gigabitEthernet 0/0.22
Siege(config-subif)# ip access-group ACL_DIRECTION in
Siege(config-subif)#exit
Siege(config)#
Siege(config)#end
Siege#write memory
%SYS-5-CONFIG_I: Configured from console by console

```

16- (1 point) Écrivez une ACL qui autorise les services WEB et DNS depuis Internet vers la DMZ sur les serveurs concernés ?

```

Siege#enable
Siege#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Siege(config)#
Siege(config)#ip access-list extended ACL_INTERNET_DMZ
Siege(config-ext-nacl)# permit udp any host 60.64.0.2 eq domain
Siege(config-ext-nacl)# permit tcp any host 60.64.0.2 eq domain
Siege(config-ext-nacl)# permit tcp any host 60.64.0.3 eq www
Siege(config-ext-nacl)# permit tcp any host 60.64.0.3 eq 443
Siege(config-ext-nacl)# permit icmp host 8.8.8.8 host 1.64.0.2 echo-reply
Siege(config-ext-nacl)# permit tcp any host 1.64.0.2 established
Siege(config-ext-nacl)# deny ip any 10.64.11.0 0.0.0.255
Siege(config-ext-nacl)# deny ip any 10.64.22.0 0.0.0.255
Siege(config-ext-nacl)# deny ip any 10.64.33.0 0.0.0.255
Siege(config-ext-nacl)# permit ip any any
Siege(config-ext-nacl)#exit
Siege(config)#
Siege(config)#interface serial 0/0/0
Siege(config-if)# ip access-group ACL_INTERNET_DMZ in
Siege(config-if)#exit
Siege(config)#
Siege(config)#end
Siege#write memory

```

17- Testez les communications depuis les VLANs IT, Direction et Employees vers Internet ?

```
C:\>ping 8.8.8.8
Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=12ms TTL=125
Reply from 8.8.8.8: bytes=32 time=10ms TTL=125
Reply from 8.8.8.8: bytes=32 time=12ms TTL=125
Reply from 8.8.8.8: bytes=32 time=16ms TTL=125

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 16ms, Average = 12ms

C:\>ping 60.64.0.2
Pinging 60.64.0.2 with 32 bytes of data:

Reply from 60.64.0.2: bytes=32 time<1ms TTL=127
Reply from 60.64.0.2: bytes=32 time<1ms TTL=127
Reply from 60.64.0.2: bytes=32 time=15ms TTL=127
Reply from 60.64.0.2: bytes=32 time=11ms TTL=127

Ping statistics for 60.64.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 15ms, Average = 6ms

C:\>ping 60.64.0.3
Pinging 60.64.0.3 with 32 bytes of data:

Reply from 60.64.0.3: bytes=32 time=1ms TTL=127
Reply from 60.64.0.3: bytes=32 time<1ms TTL=127
Reply from 60.64.0.3: bytes=32 time<1ms TTL=127
Reply from 60.64.0.3: bytes=32 time<1ms TTL=127

Ping statistics for 60.64.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Server2:

```
C:\>ping 8.8.8.8
Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=22ms TTL=125
Reply from 8.8.8.8: bytes=32 time=36ms TTL=125
Reply from 8.8.8.8: bytes=32 time=23ms TTL=125
Reply from 8.8.8.8: bytes=32 time=13ms TTL=125

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 13ms, Maximum = 36ms, Average = 23ms

C:\>ping 60.64.0.2
Pinging 60.64.0.2 with 32 bytes of data:

Reply from 60.64.0.2: bytes=32 time=9ms TTL=127
Reply from 60.64.0.2: bytes=32 time=8ms TTL=127
Reply from 60.64.0.2: bytes=32 time=7ms TTL=127
Reply from 60.64.0.2: bytes=32 time=9ms TTL=127

Ping statistics for 60.64.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 9ms, Average = 8ms

C:\>ping 60.64.0.3
Pinging 60.64.0.3 with 32 bytes of data:

Reply from 60.64.0.3: bytes=32 time=4ms TTL=127
Reply from 60.64.0.3: bytes=32 time=15ms TTL=127
Reply from 60.64.0.3: bytes=32 time=18ms TTL=127
Reply from 60.64.0.3: bytes=32 time=22ms TTL=127
```

Laptop0:

```

C:\>ping 60.64.0.2
Pinging 60.64.0.2 with 32 bytes of data:

Reply from 60.64.0.2: bytes=32 time=7ms TTL=127
Reply from 60.64.0.2: bytes=32 time=21ms TTL=127
Reply from 60.64.0.2: bytes=32 time=3ms TTL=127
Reply from 60.64.0.2: bytes=32 time=16ms TTL=127

Ping statistics for 60.64.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 21ms, Average = 11ms

C:\>ping 60.64.0.3
Pinging 60.64.0.3 with 32 bytes of data:

Reply from 60.64.0.3: bytes=32 time=18ms TTL=127
Reply from 60.64.0.3: bytes=32 time=6ms TTL=127
Reply from 60.64.0.3: bytes=32 time=13ms TTL=127
Reply from 60.64.0.3: bytes=32 time=6ms TTL=127

Ping statistics for 60.64.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 18ms, Average = 10ms

C:\>ping 8.8.8.8
Pinging 8.8.8.8 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

```

Smartphone2 :

- 18- (0,5) Connectez un nouveau client sur le VLAN IT du Switch A1. Quelle adresse IP ce client a pu obtenir ? Résolvez les conflits éventuels entre vos ACLs ? Autoriser les clients DHCP à obtenir leur adresse IP à partir du routeur ?

```
A1>enable
A1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
A1(config)#
A1(config)#interface fastEthernet 0/21
A1(config-if)# switchport mode access
A1(config-if)# switchport access vlan 33
A1(config-if)# no shutdown
A1(config-if)#exit
A1(config)#
A1(config)#end
A1#write memory
A1#
```

Config A1 :

Correction de DHCP sur Siege :

```
Siege>enable
Siege#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Siege(config)#
Siege(config)#no ip access-list extended ACL_IT
Siege(config)#
Siege(config)#ip access-list extended ACL_IT
Siege(config-ext-nacl)# permit udp any eq bootpc any eq bootps
Siege(config-ext-nacl)# permit udp any any eq bootps
Siege(config-ext-nacl)# permit ip 10.64.33.0 0.0.0.255 any
Siege(config-ext-nacl)# deny ip any any
Siege(config-ext-nacl)#exit
Siege(config)#
Siege(config)#interface gigabitEthernet 0/0.33
Siege(config-subif)# ip access-group ACL_IT in
Siege(config-subif)#exit
Siege(config)#
Siege(config)#end
Siege#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
```

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 10.64.33.23

Subnet Mask 255.255.255.0

Default Gateway 10.64.33.1

DNS Server 60.64.0.2

PC2:
Tests de ping :

C:\>ping 10.64.33.1

Pinging 10.64.33.1 with 32 bytes of data:

Reply from 10.64.33.1: bytes=32 time<1ms TTL=255
Reply from 10.64.33.1: bytes=32 time=10ms TTL=255
Reply from 10.64.33.1: bytes=32 time<1ms TTL=255
Reply from 10.64.33.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.64.33.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>ping 60.64.0.2

Pinging 60.64.0.2 with 32 bytes of data:

Reply from 60.64.0.2: bytes=32 time=7ms TTL=127
Reply from 60.64.0.2: bytes=32 time<1ms TTL=127
Reply from 60.64.0.2: bytes=32 time<1ms TTL=127
Reply from 60.64.0.2: bytes=32 time<1ms TTL=127

Ping statistics for 60.64.0.2:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 7ms, Average = 1ms

C:\>ping 60.64.0.3

Pinging 60.64.0.3 with 32 bytes of data:

Reply from 60.64.0.3: bytes=32 time=1ms TTL=127
Reply from 60.64.0.3: bytes=32 time=10ms TTL=127
Reply from 60.64.0.3: bytes=32 time<1ms TTL=127
Reply from 60.64.0.3: bytes=32 time=11ms TTL=127

Ping statistics for 60.64.0.3:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 11ms, Average = 5ms

C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time=13ms TTL=125
Reply from 8.8.8.8: bytes=32 time=3ms TTL=125
Reply from 8.8.8.8: bytes=32 time=1ms TTL=125
Reply from 8.8.8.8: bytes=32 time=8ms TTL=125

Ping statistics for 8.8.8.8:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 13ms, Average = 6ms